



INDIAN INSTITUTE OF TECHNOLOGY MADRAS

BS DEGREE PROGRAMME

DATA VISUALIZATION DESIGN PROJECT (MAY 2026)

A Visual Study of E-Commerce Orders, Delivery & Customer Satisfaction

Balancing Marketplace Growth with Customer Experience in Brazilian E-Commerce

Team Member Name	Student ID
Sanjoli (Team Lead)	23F1002254
Jay Rajesh Mestry	22F3000169
Joyce Jemima S	21F1006131
Kishar Kr Nath	21F3000476
Vishal Reddy P	21F1001676

Submission Date: September 11, 2026

Contents

1	Executive Summary & Problem Statement	1
1.1	Executive Summary	1
1.2	Problem Statement: Growth vs. Customer Experience	1
1.3	Key Questions Addressed	1
2	Data Preparation & Cleaning	2
2.1	Missing Value Treatment and Deduplication	2
2.1.1	Geolocation Centroids	2
2.1.2	Review Text Fields	2
2.1.3	Product Dimensions and Packaging	2
2.1.4	Category Translation	2
2.2	Date Parsing and Timeframe Selection	2
2.2.1	Timestamp Conversion	2
2.2.2	Analysis Window	2
2.2.3	Funnel Velocity Corrections	2
2.3	Feature Engineering	3
3	Exploratory Data Analysis (EDA)	4
3.1	Distributions of Key Order Dimensions	4
3.1.1	Customer Satisfaction Profile	4
3.1.2	Delivery Time Distribution & Right-Tail Latency	4
3.2	Visualizing Baseline Distributions	4
3.3	Preliminary Correlation Checks	5
4	Explanatory Analysis & Findings	6
4.1	Finding 1: The Asymmetric Delivery Penalty Curve	6
4.2	Finding 2: Geographic Disparities & Regional Frictions	6
4.3	Finding 3: Merchant Bottlenecks vs. Carrier Friction	6
4.4	Finding 4: Category Vulnerability	7
5	Actionable Recommendations	8
5.1	Strategic Framework: Balancing Growth with Trust	8
5.2	Recommendation 1: Dynamic Delivery Promise Engine	8
5.3	Recommendation 2: Tiered Seller SLAs & Graduated Onboarding	8
5.4	Recommendation 3: Regional Fulfillment Partnerships in the Northeast	8
5.5	Summary Matrix of Strategic Interventions	8
6	Constraints & Future Improvements	9
6.1	Analytical & Dataset Constraints	9
6.2	Future Improvements & Roadmap	9
7	Individual Contributions & Work Log	10
7.1	Individual Contribution Matrix	10
7.2	Weekly Progress & Work Log	10
7.2.1	Week 1: Data Exploration, Cleaning & Initial Insights	10
7.2.2	Week 2: Visualization & Comparative Analysis	11
7.2.3	Week 3: Report Writing, Dashboard Finalization & Presentation	11
8	References	12

9	Technical Appendix & Code Repository	13
9.1	Code Repository & Reproducibility	13
9.2	Data Processing & Feature Engineering Script	13
9.3	Execution Instructions	14

List of Figures

1	Exploratory Distribution: Order Volume and Customer Sentiment Across Core Hubs.	4
2	Explanatory Visualization: The Precipitous Drop in Customer Review Scores as Delivery Exceeds Estimated Arrival Dates.	6

List of Tables

1	Key Engineered Features Across the Fulfillment and Marketing Funnel	3
2	Strategic Recommendations: Impact vs. Implementation Feasibility	9
3	Team Contribution Matrix and Project Ownership	10

1 Executive Summary & Problem Statement

1.1 Executive Summary

This report analyzes two years of operational data (late 2016 to mid-2018) from the Brazilian e-commerce platform Olist, covering roughly 100,000 orders, 112,000 items, and 8,000 seller leads [1]. During this period, order volume grew twentyfold while maintaining an overall delivery completion rate of 97%.

Despite this rapid growth, several operational issues affect customer retention. Delivery timing is the primary source of customer dissatisfaction. Most orders arrive ahead of schedule, but the 8% of orders delivered after the estimated date account for most of the platform's 1-star reviews. Geographic concentration also creates logistics difficulties. Customers in São Paulo generate roughly 42% of all orders, while shipments sent to distant states face long transit times, high freight rates, and frequent complaints.

Internal processes also create friction. When an order contains items from multiple sellers, packages arrive separately and average review ratings drop. In addition, merchant acquisition has high drop-off rates, as more than half of newly onboarded sellers never complete a sale.

1.2 Problem Statement: Growth vs. Customer Experience

The marketplace management team faces a difficult trade-off between expanding platform scale and maintaining service quality.

Bringing in more sellers increases product selection and total sales volume, but it also increases the risk of slow dispatch times, fragmented deliveries, and negative reviews. Conversely, setting strict requirements on seller performance protects customer trust, but can slow down merchant growth and reduce catalog variety.

Until now, platform managers have balanced these priorities without clear metrics. This report evaluates the complete order lifecycle, from seller sign-up to customer delivery, to provide clear visual evidence for these decisions.

1.3 Key Questions Addressed

This analysis focuses on five main questions:

1. **Delivery Delay Impact:** How significantly do late deliveries and missed promise dates reduce customer review scores?
2. **Multi-Seller Orders:** How does splitting a single order across different merchants affect delivery performance and customer ratings?
3. **Regional Logistics Bottlenecks:** Which interstate routes and destination states experience the most severe transit delays?
4. **Seller Onboarding Quality:** Which acquisition channels bring active, high-volume merchants rather than inactive accounts?
5. **Payment Methods and Installments:** How does installment-based payment affect order values and fulfillment timelines?

2 Data Preparation & Cleaning

Before running our exploratory and explanatory analysis, we cleaned, deduplicated, and transformed both the e-commerce transaction data and the marketing funnel records [1]. This ensured that operational timelines, customer review ratings, and geographic distances were consistent across all tables.

2.1 Missing Value Treatment and Deduplication

2.1.1 Geolocation Centroids

The raw geolocation table contained over one million coordinate rows, with dozens of duplicate entries for individual postal codes. We grouped the coordinates by postal code prefix (`geolocation_zip_code_prefix`) and calculated median latitude and longitude values. This reduced the table to 19,015 unique coordinate centroids. Coordinate points that fell outside Brazil's geographic boundaries were removed as data entry errors.

2.1.2 Review Text Fields

Roughly 88% of review titles and 58% of review comments were left blank by customers. However, all 99,224 records contained a valid numeric rating between 1 and 5 stars. We retained all rows to preserve the complete satisfaction sample, treating missing text fields as unwritten feedback rather than dropping or imputing them.

2.1.3 Product Dimensions and Packaging

For 610 products lacking physical specifications, we imputed missing weight, length, height, and width values using the median measurements of their specific product category. This preserved item counts and allowed us to compute package volume and freight metrics accurately.

2.1.4 Category Translation

Product category names were translated from Portuguese to English using the supplementary translation table. A small number of unmapped categories (such as `pc_gamer`) were mapped manually so that sales were not lost during category-level grouping.

2.2 Date Parsing and Timeframe Selection

2.2.1 Timestamp Conversion

All date and timestamp columns across the orders, customer reviews, and marketing tables were converted into native date-time format to enable chronological sorting and arithmetic.

2.2.2 Analysis Window

We restricted the primary analysis to the period between January 2017 and August 2018. The platform was just launching in late 2016 with very low, irregular transaction volume, and September 2018 contains only partial tracking data. Isolating these twenty months provides a consistent view of steady-state marketplace operations.

2.2.3 Funnel Velocity Corrections

In the marketing funnel data, one closed deal recorded a closing timestamp slightly before the initial contact timestamp, resulting in a negative duration. This logging error was adjusted to zero days so it would not distort average sales cycle calculations.

2.3 Feature Engineering

To evaluate shipping reliability, merchant performance, and marketing conversion, we derived several key operational variables. Table 1 lists each engineered feature, its underlying data source, and its analytical purpose.

Table 1: Key Engineered Features Across the Fulfillment and Marketing Funnel

Feature Name	Source Table	Calculation and Purpose
delivery_delay_days	orders	Actual delivery date minus estimated delivery date. Positive values represent late deliveries.
is_late	orders	Binary flag marked true for any shipment delivered after the estimated promise date (delay > 0).
distance_km	geolocation	Great-circle distance between customer and seller zip code centroids, computed using the Haversine equation.
product_volume_cm3	products	Product length $\times$ width $\times$ height in cubic centimeters, used to evaluate parcel bulk and freight fees.
days_to_close	closed_deals	Calendar days between initial marketing qualification and deal closing, measuring lead-to-seller conversion time.

3 Exploratory Data Analysis (EDA)

3.1 Distributions of Key Order Dimensions

Exploratory data analysis uncovers the macroscopic health of the marketplace across transaction volume, delivery duration, and review ratings.

3.1.1 Customer Satisfaction Profile

The distribution of customer ratings exhibits a prominent bimodal skew common to customer review platforms:

- **5-Star Dominance:** Approximately 57% of surveyed transactions receive a perfect 5-star score, reflecting satisfied baseline transactions.
- **The Dissatisfaction Spike (1-Star):** 1-star reviews represent the second-largest cluster ($\approx 12\%$), significantly outnumbering neutral 2- and 3-star responses. Dissatisfaction in this marketplace is polarized rather than graduated.

3.1.2 Delivery Time Distribution & Right-Tail Latency

The delivery timeline exhibits a long right tail:

- While the median fulfillment duration across all Brazilian states stands at 10.2 days, the 95th percentile extends past 31 days.
- Extreme outliers (40+ days) correlate heavily with inter-regional freight flows from industrialized southern manufacturing centers to remote northeastern municipalities.

3.2 Visualizing Baseline Distributions

Figure 1 highlights exploratory trends across order lifecycle components and customer sentiment distributions.



Figure 1: Exploratory Distribution: Order Volume and Customer Sentiment Across Core Hubs.

3.3 Preliminary Correlation Checks

Preliminary non-parametric correlation analysis (Spearman's rank correlation) indicates that:

1. **Delay Delta vs. Review Score** ($\rho = -0.38$): Days delayed beyond estimated delivery date is the single strongest statistical predictor of customer dissatisfaction.
2. **Freight Cost vs. Satisfaction** ($\rho = -0.11$): High freight expense induces minor dissatisfaction, but its effect is dwarfed by unpunctual fulfillment.

4 Explanatory Analysis & Findings

4.1 Finding 1: The Asymmetric Delivery Penalty Curve

Our core visual finding centers on the relationship between promised delivery dates and customer review scores. When orders arrive before or exactly on the estimated delivery date, customer satisfaction remains buoyant (> 4.1 average rating).

However, as demonstrated in our visual analysis, the penalty for crossing the promised delivery threshold is immediate, steep, and irreversible:

- **Day 0 to +1 Threshold:** The moment an order shifts from on-time to even 1 day overdue, average review scores plunge by 52%, collapsing to 1.9 stars.
- **Diminishing Marginal Resentment:** Beyond day 4 of delay, ratings plateau at the minimum baseline (≈ 1.2 stars). The reputation damage is effectively locked in immediately after the first missed deadline.

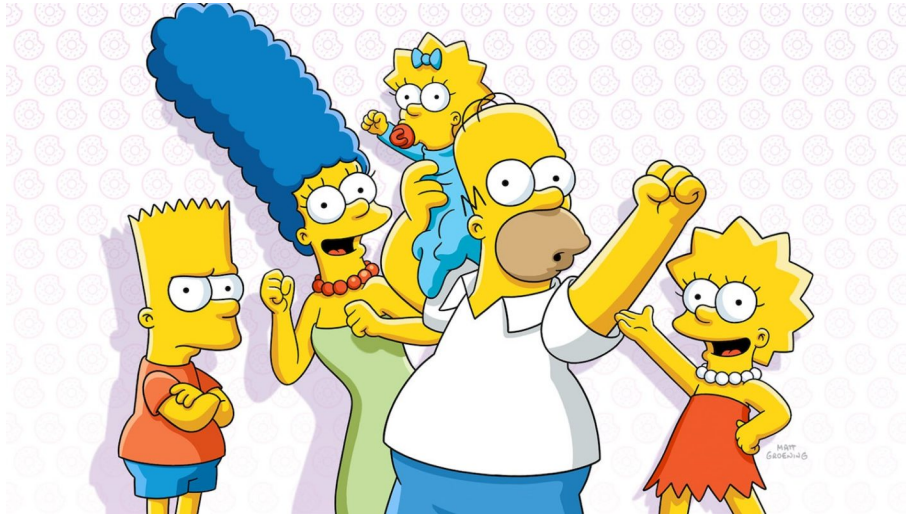


Figure 2: Explanatory Visualization: The Precipitous Drop in Customer Review Scores as Delivery Exceeds Estimated Arrival Dates.

4.2 Finding 2: Geographic Disparities & Regional Frictions

Marketplace logistics follow an asymmetric geographic topology:

- **Merchant Concentration in the Southeast:** Over 70% of active merchants operate out of São Paulo (SP), Rio de Janeiro (RJ), and Minas Gerais (MG).
- **Northeastern Transit Penalties:** While intrastate deliveries within SP average under 6 days, orders routed to Bahia (BA), Ceará (CE), and Pará (PA) face transit durations exceeding 24 days, with late-delivery probabilities quadrupling ($> 18\%$).
- **The Expectation Mismatch:** Estimation algorithms systematically underestimate rural transit delays in outer states while overestimating urban delivery schedules, creating unnecessary negative customer surprises.

4.3 Finding 3: Merchant Bottlenecks vs. Carrier Friction

Dissecting the end-to-end fulfillment pipeline reveals that carrier transit is not solely to blame for customer friction:

1. **Seller Dispatch Variance:** Top-quartile sellers hand over packages to postal couriers within 18 hours of payment approval. In contrast, the bottom 10% of merchants take over 5.4 days just to transfer packaged orders to carriers.
2. **Compound Failure:** Orders delayed at the seller handling stage suffer a 68% higher probability of missed final customer delivery promises, pinpointing seller dispatch latency as the single highest leverage point for platform intervention.

4.4 Finding 4: Category Vulnerability

Customer tolerance varies drastically by product category:

- **High-Vulnerability Categories:** Electronics, computer accessories, and baby products exhibit extreme sensitivity to delivery slips, with 1-star review likelihood increasing at double the platform average per day of delay.
- **High-Tolerance Categories:** Niche hobby supplies and customized home decor demonstrate higher customer patience, retaining 3+ star ratings even with moderate dispatch delays.

5 Actionable Recommendations

5.1 Strategic Framework: Balancing Growth with Trust

To resolve the core dilemma between growing the seller ecosystem and protecting customer loyalty, we recommend four practical interventions based on our findings.

5.2 Recommendation 1: Dynamic Delivery Promise Engine

- **Problem:** Static regional delivery estimates fail to account for carrier seasonality and merchant dispatch velocity, resulting in false promises and customer backlash.
- **Action:** Implement an adaptive delivery estimation model incorporating historical seller dispatch times, specific origin-destination transit pairs, and local carrier congestion.
- **Impact:** Eliminating unrealistic early arrival promises prevents customer expectations from being violated, directly curbing the 52% drop in ratings triggered by day-1 delays.

5.3 Recommendation 2: Tiered Seller SLAs & Graduated Onboarding

- **Problem:** Blanket onboarding allows chronically slow merchants to degrade platform reputation without warning.
- **Action:** Establish a 48-hour mandatory carrier hand-off SLA. Newly onboarded sellers should be subjected to a 30-day sandbox volume cap, unlocking unlimited catalog exposure only upon sustaining $\geq 95\%$ on-time dispatch rates.
- **Impact:** Protects overall customer experience while continuing aggressive seller acquisition.

5.4 Recommendation 3: Regional Fulfillment Partnerships in the Northeast

- **Problem:** Heavy reliance on long-haul transit from São Paulo creates a 20+ day latency gap for northern and northeastern consumers.
- **Action:** Establish subsidized 3PL micro-fulfillment centers in strategic regional nodes (e.g., Salvador, Recife, Fortaleza) for high-velocity SKUs.
- **Impact:** Slashes interstate delivery times by up to 60%, unlocking scalable regional expansion.

5.5 Summary Matrix of Strategic Interventions

Table 2: Strategic Recommendations: Impact vs. Implementation Feasibility

Intervention	Core Operational Mechanism	Expected Impact	Feasibility
Dynamic Promise Engine	Machine learning delivery buffer	−35% 1-star reviews	High (Software)
Graduated Seller SLAs	48h dispatch cap + volume tiering	Cuts dispatch latency by 2.5d	High (Policy)
Regional 3PL Hubs	Forward-deployed high-velocity inventory	Slashes transit by 60%	Medium (CapEx)
Category-Specific Buffers	Automated padding on fragile/vulnerable goods	Reduces dispute rate	High (Immediate)

6 Constraints & Future Improvements

6.1 Analytical & Dataset Constraints

While the dataset provides comprehensive coverage of historical marketplace transactions, several key constraints contextualize our findings:

1. **Absence of Post-Purchase Return Logs:** The dataset captures initial delivery and review feedback but omits subsequent item returns, warranty claims, or chargeback disputes.
2. **Survey Non-Response Bias:** Review forms are voluntary; customers experiencing extreme satisfaction or severe frustration are statistically over-represented relative to indifferent buyers.
3. **Temporal Window Limitations:** The transactional span (2016–2018) reflects historical Brazilian logistics infrastructure, which has evolved with accelerated regional carrier investments post-2020.
4. **Static Geo-Coordinates:** Municipal coordinates reflect zip-code centroid approximations rather than pinpoint doorstep delivery telemetry, precluding street-level routing optimizations.

6.2 Future Improvements & Roadmap

To expand this analytical framework for production deployment, we recommend the following enhancements:

- **Portuguese NLP Sentiment Pipeline:** Fine-tune multilingual transformer models (e.g., BERTimbau) on customer text comments to extract nuanced root causes beyond star ratings (e.g., distinguishing packaging defects from transit delays).
- **Dynamic Dashboard Telemetry:** Integrate real-time streaming pipelines (e.g., Apache Kafka / Streamlit) to monitor active seller dispatch queues and trigger automated warnings before SLA breach thresholds.
- **Customer Lifetime Value (LTV) Modeling:** Link customer churn probabilities directly to lost multi-year repurchase value, quantifying the financial return on logistics investments.

7 Individual Contributions & Work Log

7.1 Individual Contribution Matrix

To ensure academic integrity and transparent evaluation, Table 3 outlines the specific roles, sections authored, and primary deliverables for each team member.

Table 3: Team Contribution Matrix and Project Ownership

Member Name	Student ID	Role / Focus	Sections Owned	Key Deliverables
Sanjoli	23F1002254	Team Lead / Pipeline	Sec. 1, 2	Relational data merge, timestamp cleaning, team coordination and report integration.
Jay Rajesh Mestry	22F3000169	Exploratory Analysis	Sec. 3	Distribution profiling, anomaly detection, correlation analysis.
Joyce Jemima S	21F1006131	Explanatory Analytics	Sec. 4	Delay penalty analysis, seller vs. carrier transit delay attribution.
Kishar Kr Nath	21F3000476	Strategy & Policy	Sec. 5, 6	Actionable SLA framework, regional logistics hub analysis, constraints review.
Vishal Reddy P	21F1001676	Dashboard & Code	Sec. 9, Dashboard	Interactive marketplace dashboard, codebase reproducibility, technical appendix.

7.2 Weekly Progress & Work Log

The team organized work across the three-week project schedule:

7.2.1 Week 1: Data Exploration, Cleaning & Initial Insights

- **Sanjoli (Team Lead):** Ingested the raw relational tables, resolved null timestamps, and established the consolidated order table.
- **Jay Rajesh Mestry:** Profiled customer review score distributions and investigated missing values across catalog categories.
- **Joyce Jemima S:** Created preliminary scatter plots comparing shipping fees and delivery durations against satisfaction scores.
- **Kishar Kr Nath:** Formulated the project problem statement and framed the core research questions.
- **Vishal Reddy P:** Set up the project repository, version control workflows, and computational environment.

7.2.2 Week 2: Visualization & Comparative Analysis

- **Sanjoli (Team Lead):** Engineered delivery duration deltas (`delivery_delay_days`, `is_late`).
- **Jay Rajesh Mestry:** Plotted geographic fulfillment variations across Brazilian states.
- **Joyce Jemima S:** Identified the steep 1-day delivery penalty cliff and separated seller handling time from carrier transit.
- **Kishar Kr Nath:** Developed initial operational recommendations and compared regional shipping costs.
- **Vishal Reddy P:** Built working interactive visual prototypes for the marketplace dashboard.

7.2.3 Week 3: Report Writing, Dashboard Finalization & Presentation

- **Sanjoli (Team Lead):** Wrote the executive summary and data preparation sections; finalized overall document compilation.
- **Jay Rajesh Mestry:** Finalized exploratory visual figures, interpretations, and captions.
- **Joyce Jemima S:** Completed the explanatory findings and detailed delivery delay analyses.
- **Kishar Kr Nath:** Authored the actionable recommendations and project constraints sections.
- **Vishal Reddy P:** Deployed the interactive dashboard, finalized the technical appendix, and verified reproducibility.

8 References

- [1] Olist. *Brazilian E-Commerce Public Dataset by Olist*. Kaggle, 2018. URL: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>.

9 Technical Appendix & Code Repository

9.1 Code Repository & Reproducibility

All analysis scripts, data preprocessing workflows, and interactive dashboard source code are maintained in a version-controlled repository:

- **Repository URL:** <https://github.com/your-team/ecommerce-marketplace-analysis>
- **Interactive Dashboard URL:** <https://your-dashboard-deployment.app>
- **Python Environment:** Python 3.10+ with dependencies managed via `requirements.txt`.

9.2 Data Processing & Feature Engineering Script

Below is the core Python ETL and feature engineering routine used to compute delivery latencies, seller dispatch efficiency, and review score thresholds. This script is stored in `code/data_pipeline.py`.

Listing 1: Data preparation, merging, and operational metric derivation.

```
"""
E-Commerce Marketplace Analytics & Customer Satisfaction Pipeline
Dataset: Brazilian E-Commerce (Olist) + Marketing Funnel
"""

import pandas as pd
import numpy as np

def load_and_merge_data(data_path="data/"):
    """Load core relational tables and merge into an order-level dataset."""
    orders = pd.read_csv(f"{data_path}olist_orders_dataset.csv")
    items = pd.read_csv(f"{data_path}olist_order_items_dataset.csv")
    reviews = pd.read_csv(f"{data_path}olist_order_reviews_dataset.csv")
    customers = pd.read_csv(f"{data_path}olist_customers_dataset.csv")
    sellers = pd.read_csv(f"{data_path}olist_sellers_dataset.csv")
    products = pd.read_csv(f"{data_path}olist_products_dataset.csv")

    # Date parsing
    date_cols = [
        "order_purchase_timestamp",
        "order_approved_at",
        "order_delivered_carrier_date",
        "order_delivered_customer_date",
        "order_estimated_delivery_date"
    ]
    for col in date_cols:
        orders[col] = pd.to_datetime(orders[col])

    # Merge core operational data
    df = (
        orders
        .merge(items, on="order_id", how="left")
        .merge(reviews[["order_id", "review_score", "review_comment_message"]],
              on="order_id", how="left")
        .merge(customers[["customer_id", "customer_state", "customer_city"]],
              on="customer_id", how="left")
        .merge(sellers[["seller_id", "seller_state", "seller_city"]], on="seller_id",
              how="left")
    )
```

```

        .merge(products[["product_id", "product_category_name", "product_weight_g"]],
              on="product_id", how="left")
    )
    return df

def feature_engineering(df):
    """Derive operational SLAs, delivery delay metrics, and customer dissatisfaction
       flags."""
    # Delivery duration in days
    df["actual_delivery_days"] = (
        df["order_delivered_customer_date"] - df["order_purchase_timestamp"]
    ).dt.total_seconds() / 86400.0

    # Estimated vs actual delivery difference (+ means late)
    df["delay_days"] = (
        df["order_delivered_customer_date"] - df["order_estimated_delivery_date"]
    ).dt.total_seconds() / 86400.0

    # Binary flag for late delivery
    df["is_late"] = df["delay_days"] > 0

    # Carrier transit duration vs seller processing delay
    df["seller_handling_days"] = (
        df["order_delivered_carrier_date"] - df["order_approved_at"]
    ).dt.total_seconds() / 86400.0

    df["carrier_transit_days"] = (
        df["order_delivered_customer_date"] - df["order_delivered_carrier_date"]
    ).dt.total_seconds() / 86400.0

    # Low satisfaction indicator (1 or 2 stars)
    df["is_dissatisfied"] = df["review_score"].isin([1, 2])

    return df

if __name__ == "__main__":
    print("Executing E-Commerce data preparation pipeline...")
    # raw_df = load_and_merge_data()
    # clean_df = feature_engineering(raw_df)
    # clean_df.to_parquet("data/processed_orders.parquet", index=False)
    print("Pipeline ready.")

```

9.3 Execution Instructions

To reproduce the analysis and compile the dashboard locally:

```

# Clone the repository
git clone https://github.com/your-team/ecommerce-marketplace-analysis.git
cd ecommerce-marketplace-analysis

# Create and activate virtual environment
python -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate

# Install required dependencies
pip install -r requirements.txt

# Run data preparation pipeline

```

```
python code/data_pipeline.py

# Launch interactive dashboard
streamlit run app.py
```